Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 23

Proof by Contrapositive

Covers Chartrand & Zhang, Section 3.3

Status of this set

Ungraded practice. Solutions released with it. The midterm is Wednesday, October 14 — this set is also good review of proof writing.

Part I: How to Practice This Material

Say which implication you are proving

The template

To prove $P \Rightarrow Q$ by contrapositive:

1. **Write down the contrapositive** $(\sim Q) \Rightarrow (\sim P)$ explicitly.
2. **Assume** $\sim Q$ and give a direct proof of $\sim P$.
3. **Conclude** by citing that an implication and its contrapositive are equivalent.

Step 1 is not optional. Writing it down is what keeps you from proving the converse by accident.

When to reach for it

Use the contrapositive when the **hypothesis is hard to use** and the **conclusion is easy to negate**. Typical trigger: the hypothesis involves n^2 or a product, while the conclusion is a simple parity statement.
Negating swaps them, and the easy one becomes your starting point.

Three mistakes to expect

1. **Proving the converse instead.** Negate *and* swap.
2. **Assuming $\sim P$ and deriving $\sim Q$.** That is the inverse, which is not equivalent.
3. **Not stating which implication is being proved.** Correct algebra on the wrong claim earns nothing.

Part II: Practice Problems

Section A — Write the contrapositive first

1. For each, write the contrapositive, then prove the original statement.

(a) Let $n \in \mathbb{Z}$. If $7n - 3$ is even, then n is odd.

(b) Let $n \in \mathbb{Z}$. If $n^2 - 6n + 5$ is odd, then n is even.

Section B — Choosing the method

2. For each statement, say whether a direct proof or a contrapositive proof is more practical, and why. Then prove it.

(a) If n is even, then $9n - 6$ is even.

(b) If n^3 is even, then n is even.

Section C — Biconditionals

3. Prove: let $n \in \mathbb{Z}$. Then $5n - 2$ is odd if and only if n is odd.

Section D — Finding the flaw

4. A student offers this proof. Identify what is wrong.

Claim: let $n \in \mathbb{Z}$. If $3n + 4$ is odd, then n is odd.

Proof. Assume n is odd. Then $n = 2k + 1$, so $3n + 4 = 6k + 7 = 2(3k + 3) + 1$, which is odd. Hence the claim holds.

Section E — Writing

5. In five to eight complete sentences, explain why proving the contrapositive proves the original statement, while proving the converse does not. Name the results from earlier in the term that settle each, and give a concrete pair illustrating the difference.

Section F — Challenge (optional)

6. Prove: let $m, n \in \mathbb{Z}$. If mn is odd, then m and n are both odd.